

AMF Windows sample001 — Memory-Triage Forensic Report

🟡 OFFICIAL — SENSITIVE (DFIR / training case) · Live Agentropix-SIFT memory triage

Volatility 3 triage of a 511 MiB raw Windows XP RAM image. One medium-severity finding approved and sealed: 15 RWX injected-code regions recovered by `malfind`, concentrated in `winlogon.exe`.

Field	Value
Report ID	3c5261e7abc4fb7de891e0ee4347ead2519d6414b16cef8198a43dcb2347e634
Case ID	AMF-WIN-SAMPLE001
Case name	AMF Windows sample001 (Art of Memory Forensics)
Examiner	victor.galvan
Profile	full
Incident type	dfir
Severity	🟡 Medium
Snapshot at (UTC)	2026-06-06T23:17:52Z
Generated at (UTC)	2026-06-07
MCP host	<TAILNET-HOST>
Seal	hmac-sha256:29479f98... (1 approved finding sealed)

Audiences served: CISO / stakeholder · SOC / blue team · Red team · Audit.

1. Executive Summary

A 511 MiB raw Windows XP memory image (`sample001.bin`) was registered, hashed (SHA-256 `03242077...`) and triaged through the Agentropix-SIFT MCP server using Volatility 3. The triage recovered a coherent system snapshot — 21 processes, 229 services, a clean two-root process forest with zero orphans and zero LOLBin flags — alongside 15 RWX (`PAGE_EXECUTE_READWRITE`) memory regions surfaced by `malfind` , heavily concentrated in `winlogon.exe` (10 of 15). One medium-severity finding (`F-AMF-S001-001` , MITRE T1055 Process Injection) was persisted as a DRAFT and carried through the examiner approval loop into a sealed report. No live network sockets were recovered (an honest empty result on this XP-era image).

Key finding —  `F-AMF-S001-001` : 15 RWX injected-code regions recovered by `malfind` , concentrated in `winlogon.exe` (×10) — MITRE T1055 Process Injection, medium severity, approved and HMAC-sealed.

2. KPI Summary

KPI	Value	Detail
Approved findings	1	medium × 1 (<code>F-AMF-S001-001</code> , sealed)
Hosts in scope	1	<code>sample001</code> (Windows XP RAM image)
IOCs catalogued	0	no IOCs promoted — RWX VADs are host artefacts, not externally-actionable indicators
Attacker dwell	Indeterminate	single RAM snapshot, no longitudinal data
MITRE techniques	1	T1055 (Process Injection)
Initial access	Not determined	RAM-only snapshot — no initial-access vector recoverable

Host roster: `sample001` — single Windows XP host, raw memory image

`/cases/AMF_MemorySamples/windows/sample001.bin` .

3. Risk Matrix

The single approved finding (F-AMF-S001-001) is scored Possible × Moderate. RWX regions in winlogon.exe are a strong injection indicator, but on this XP-era training image they cannot be conclusively distinguished from legitimate JIT/unpacked code without a follow-up dump-and-disassemble, so likelihood is held at Possible and impact at Moderate.

Impact ↓ / Likelihood →	1 Rare	2 Unlikely	3 Possible	4 Likely	5 Almost certain
5 Severe	.	.	.	.	.
4 Major	.	.	.	.	.
3 Moderate	.	.	● F1	.	.
2 Minor	.	.	.	.	.
1 Negligible	.	.	.	.	.

Scored findings

Ref	Risk	Impact	Likelihood	Score	Severity
F1	F-AMF-S001-001 — RWX injected-code regions in winlogon.exe	3	3	9	● Medium

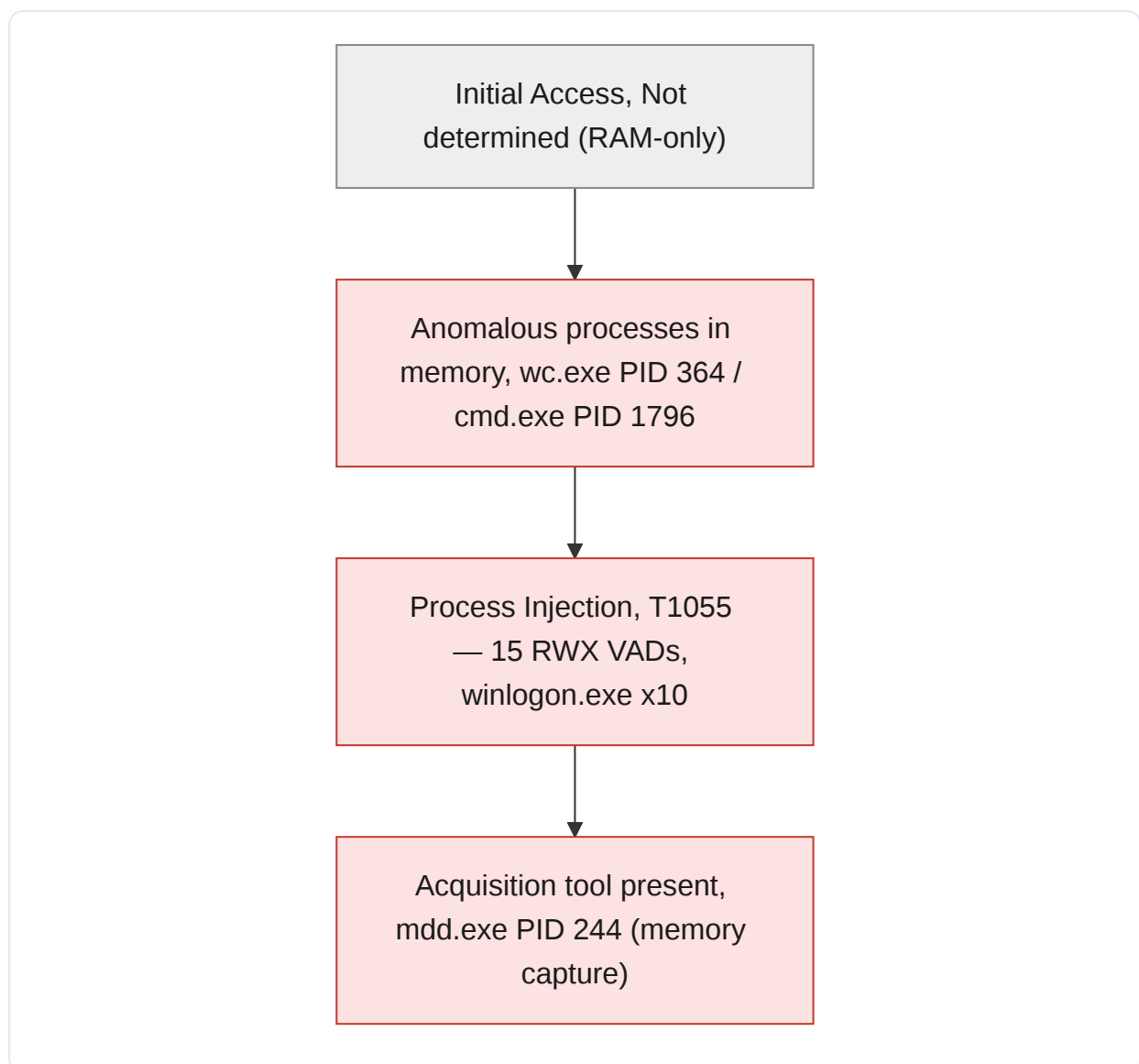
4. Key Findings

- F-AMF-S001-001 — 15 RWX (PAGE_EXECUTE_READWRITE) injected-code regions recovered by malfind, concentrated in winlogon.exe (×10) (Medium) — strongest real signal in the image; persisted, examiner-approved, HMAC-sealed (T1055 Process Injection).

5. Attack Chain & MITRE ATT&CK

This is a single RAM snapshot, so a full kill chain cannot be reconstructed. The chain below reflects only what is directly evidenced in memory: anomalous late-spawning processes (`wc.exe` , `cmd.exe` , the `mdd.exe` acquisition tool) and the RWX-region injection footprint that became the approved finding. Initial access and C2 stages are not determined — no network sockets were recovered.

5.1 Attack chain



5.2 MITRE ATT&CK techniques

Tactic	Technique ID	Technique	Evidence / how observed
Defense Evasion / Privilege Escalation	T1055	Process Injection	15 PAGE_EXECUTE_READWRITE VADs recovered by <code>get_malfind</code> — winlogon.exe x10, lsass.exe x2, csrss.exe x1, msmsgs.exe x1, msimn.exe x1. Basis of approved finding F-AMF-S001-001 .
Discovery	T1057	Process Discovery	<code>get_pslist / cmdline</code> enumerated 21 processes (dry-run finding amf-win-s001-001 validated this technique mapping, not persisted).

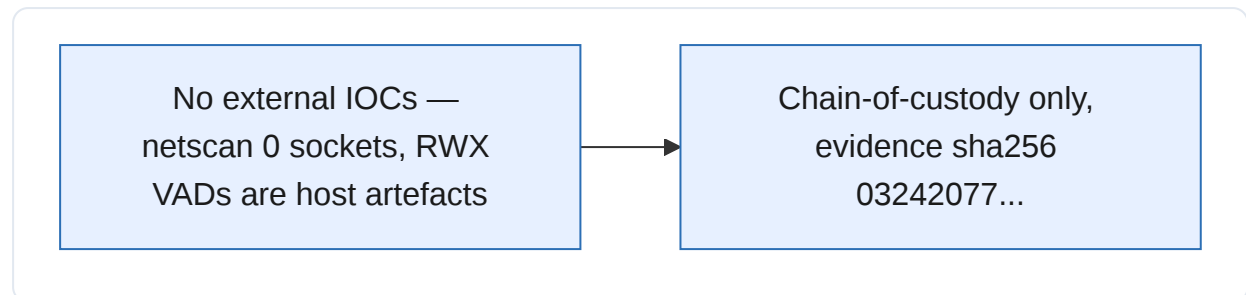
6. IOC Catalogue

No externally-actionable IOCs were promoted for this case. The recovered RWX VAD regions are host-internal memory artefacts rather than network/file indicators, and `get_netscan` returned zero sockets, so there were no addresses, domains, or external hashes to catalogue. The evidence-image SHA-256 below is recorded for chain-of-custody, not as a threat indicator.

Type	Value	Role	Confidence	MITRE	Source
—	—	—	—	—	No IOCs promoted — RWX VADs are host artefacts, <code>netscan</code> returned 0 sockets

Negative space (surveyed, not observed): `get_netscan` returned 0 sockets (no C2 / remote endpoints); `build_process_tree` flagged 0 suspicious / 0 LOLBin processes and 0 orphans; no external file hashes or domains surfaced.

6.1 IOC provenance



7. Host Artefacts

Single host: `sample001`. The process table is shown in full (21 processes from `get_pslist` / `cmdline`); the 229-service inventory from `get_svcscan` is summarised with representative rows. The `malfind` RWX footprint is summarised in the per-process counts in §5.2 rather than repeated as a table. Create-times are the in-memory process creation timestamps from the XP image (2012-11-26/27 UTC).

7.1 Host — sample001

Processes (pslist + cmdline)

PID	PPID	Image	Started (UTC)	Threads	Command line	Source
4	0	System	N/A	51	(none)	get_pslist
356	4	smss.exe	2012-11-26 22:03:28	3	\SystemRoot\System32\smss.exe	get_pslist / cmdline
604	356	csrss.exe	2012-11-26 22:03:29	12	C:\WINDOWS\system32\csrss.exe ObjectDirectory=\Windows...	cmdline
628	356	winlogon.exe	2012-11-26 22:03:29	18	winlogon.exe	cmdline
680	628	services.exe	2012-11-26 22:03:30	15	C:\WINDOWS\system32\services.exe	cmdline
692	628	lsass.exe	2012-11-26 22:03:30	22	C:\WINDOWS\system32\lsass.exe	cmdline
852	680	svchost.exe	2012-11-26 22:03:31	14	svchost -k DcomLaunch	cmdline
940	680	svchost.exe	2012-11-26 22:03:31	9	svchost -k rpcss	cmdline
1024	680	svchost.exe	2012-11-26 22:03:32	76	svchost.exe -k netsvc	cmdline
1068	680	svchost.exe	2012-11-26 22:03:32	5	svchost.exe -k NetworkService	cmdline
1116	680	svchost.exe	2012-11-26 22:03:33	14	svchost.exe -k LocalService	cmdline

PID	PPID	Image	Started (UTC)	Threads	Command line	Source
1348	680	spoolsv.exe	2012-11-26 22:03:34	10	C:\WINDOWS\system32\spoolsv.exe	cmdline
1888	680	alg.exe	2012-11-26 22:03:35	6	C:\WINDOWS\System32\alg.exe	cmdline
284	244	explorer.exe	2012-11-26 22:03:58	9	C:\WINDOWS\Explorer.EXE	cmdline
548	284	msmsgs.exe	2012-11-26 22:04:03	3	"C:\Program Files\Messenger\msmsgs.exe" /background	cmdline
556	284	ctfmon.exe	2012-11-26 22:04:03	1	"C:\WINDOWS\system32\ctfmon.exe"	cmdline
1628	1024	wuauclt.exe	2012-11-26 22:04:43	3	"C:\WINDOWS\system32\wuauclt.exe"	cmdline
1984	284	msimn.exe	2012-11-26 22:06:33	7	"C:\Program Files\Outlook Express\msimn.exe"	cmdline
364	1024	wc.exe	2012-11-27 01:30:00	1	wc.exe -e -o h.out	cmdline
1796	284	cmd.exe	2012-11-27 01:56:21	1	"C:\WINDOWS\system32\cmd.exe"	cmdline
244	1796	mdd.exe	2012-11-27 01:57:28	1	mdd.exe -o callb-memdump.bin	cmdline

Analyst note — three processes spawn ~3.5 hours after boot: `wc.exe -e -o h.out` (PID 364, child of `svchost -k netsvcs`), `cmd.exe` (PID 1796), and `mdd.exe -o callb-memdump.bin` (PID 244) — `mdd.exe` is the memory-acquisition tool that captured this very image. The `wc.exe` under `svchost` and its non-standard `-e -o h.out` arguments are the most analyst-worthy anomaly. `explorer.exe` PPID (244) referencing the reused `mdd.exe` PID is a known AMF PID-reuse artefact, not a live parentage edge.

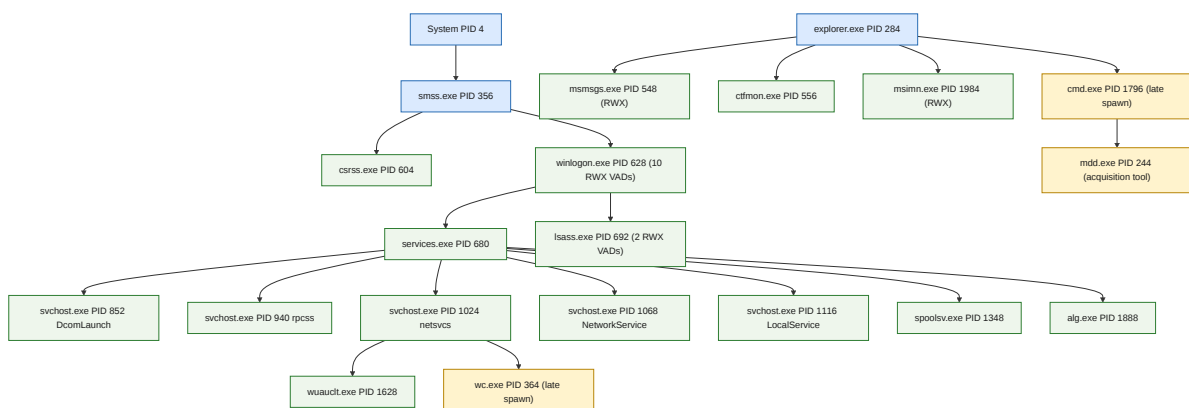
Services (svcsan — 229 total, representative rows)

Service	State	Class	Source
ACPI	SERVICE_RUNNING	SERVICE_KERNEL_DRIVER	get_svcsan
AFD	SERVICE_RUNNING	SERVICE_KERNEL_DRIVER	get_svcsan
ALG	SERVICE_RUNNING	SERVICE_WIN32_OWN_PROCESS	get_svcsan
AudioSrv	SERVICE_RUNNING	SERVICE_WIN32_SHARE_PROCESS	get_svcsan
... 225 more	—	—	229 services total

malfind (RWX VADs): 15 `PAGE_EXECUTE_READWRITE` regions — `winlogon.exe` x10 (PID 628), `lsass.exe` x2 (692), `csrss.exe` x1 (604), `msmsgs.exe` x1 (548), `msimn.exe` x1 (1984).

7.2 Process Tree

`build_process_tree` reported a clean forest: 21 processes, 2 roots, 0 orphans, 0 suspicious / LOLBin flags. Shown below from PID/PPID edges.



The two reported roots are `System` (PID 4) and `smss.exe` (PID 356). `explorer.exe` (PID 284) renders as a second visual subtree because its in-memory PPID (244) points at the reused `mdm.exe` PID — an AMF artefact, hence drawn as an independent root rather than an inferred edge.

8. Network Artefacts

`get_netscan` recovered 0 sockets from this image. On an XP-era raw dump this is an honest empty result — either no network state was resident at capture time, or the pool tags were not recoverable. No remote endpoints, listeners, or C2 channels are evidenced.

Process (PID)	Local	Remote	State	Purpose	Source
—	—	—	—	—	<code>get_netscan</code> returned 0 sockets (no network state recovered)

No further network artefacts (DNS cache, connections, listeners) were recoverable from this snapshot.

9. Detailed Findings

9.1  `F-AMF-S001-001` — 15 RWX (`PAGE_EXECUTE_READWRITE`) injected-code regions recovered by `malfind`, concentrated in `winlogon.exe` (×10)

Severity	 Medium (confidence 0.7)
Host	sample001
Technique	T1055 — Process Injection
Status	APPROVED (sealed) — <code>hmac-sha256:29479f98...</code>

`get_malfind` recovered 15 memory regions marked `PAGE_EXECUTE_READWRITE` across five processes, with a pronounced concentration in `winlogon.exe` (PID 628, 10 of 15 regions). RWX VADs are a primary signature of code injection / unpacking because legitimate code is normally execute-read, not execute-read-write. The finding was first validated under `dry_run:true` (anti-hallucination gate, `indexed:false`), then persisted under a single-scope evidence-gate mutation token (`scope=index_findings` , 30-min TTL) as a DRAFT, and finally transitioned DRAFT → APPROVED through the Examiner Portal. The malfind pass exceeded the default 180 s SDK request bound and was re-run standalone with a 300 s `callTool` timeout, completing in 75 s.

Evidence — evidence `f2649687...` (image `sample001.bin`) · sha256 `03242077eb3364fb248d1c7730fd0a94074583df8deb2606d6c93c31316d561c` · capture: `get_malfind` (volatility3.windows.malfind.Malfind), raw_stdout_sha256 of cmdline cross-check `d63cc770...` .

Approval caveat (audit) — the DRAFT → APPROVED transition was performed by Playwright demo automation (`examiner_id=victor.galvan` , reason "SIMULATED examiner approval (demo only)"), NOT a human HMAC sign-off. Approval ID `4a881577...` , approved_at 2026-06-06T23:17:43Z. In a real case this transition is a Hard-Stop requiring a human examiner's HMAC challenge-response.

10. Timeline

The timeline below is reconstructed from in-memory process creation timestamps (XP image clock, 2012) and the Agentropix triage run clock (2026). The early 2012-11-26 cluster is normal boot sequencing; the late 2012-11-27 01:30–01:57 cluster (`wc.exe` , `cmd.exe` , `mdd.exe`) is the analyst-relevant activity window, ending with the memory acquisition itself.

Time (UTC)	Host	Event	Technique	Source
2012-11-26 22:03:28	sample001	Boot sequence begins (smss.exe → csrss/winlogon → services/lsass)	—	pslist create_time
2012-11-26 22:06:33	sample001	User session steady state (explorer, msmsgs, msimn)	—	pslist create_time
2012-11-27 01:30:00	sample001	wc.exe -e -o h.o ut spawned under svchost netsvcs (PID 364)	—	pslist / cmdline
2012-11-27 01:56:21	sample001	cmd.exe opened (PID 1796)	—	pslist / cmdline
2012-11-27 01:57:28	sample001	mdd.exe -o callb -memdump.bin runs — memory acquisition (PID 244)	—	pslist / cmdline
2026-06-06 23:17:43	sample001	Finding F-AMF-S001-001 APPROVED (demo automation)	T1055	approval 4a881577...
2026-06-06 23:17:52	sample001	Sealed report generated (3c5261e7... , approved_finding_count 1)	—	report_generate(full)

11. Agentropix Performance

The full §3.A manual sequence ran live against the MCP server on 2026-06-06. The heaviest stage was `get_malfind` (re-run at a 300 s timeout, completed in 75 s). All other tools returned within the default SDK request bound.

Metric	Value	Detail
Run time	~2.7 hrs wall (20:36 → 23:18 UTC)	includes manual examiner-portal step and demo setup
MCP tool calls	11 steps / 16+ tool invocations	health, case_init, case_status, evidence_register, get_image_info, pslist, netscan, malfind, svcscan, build_process_tree, cmdline, record_finding ×2, approval, report_generate
Evidence size	536,330,240 bytes (≈ 511 MiB)	raw <code>.bin</code> Windows XP RAM image
Disk recall	72/72 (100%)	canonical (.crew/facts.md)
Memory recall	108/118 (91.5%)	canonical (.crew/facts.md)
Test suite	4464	canonical (.crew/facts.md)

Per-stage timing

Stage	Capture	Duration	Result
Health probe	<code>health</code>	< 1 s	72 tools registered
Evidence register + hash	<code>evidence_register</code>	seconds (511 MiB hash)	sha256 <code>03242077...</code>
Process / service triage	<code>pslist · netscan · svcscan · build_process_tree</code>	within 180 s bound	21 procs, 0 sockets, 229 svcs, clean tree
Injected-code scan	<code>get_malfind</code> (300 s timeout)	75 s	15 RWX hits
Report generation	<code>report_generate(full)</code>	seconds	report <code>3c5261e7...</code> , 1 approved finding

12. Coverage Attestation

This report is grounded entirely on real captured MCP output. Numeric recall / test figures are canonical (`.crew/facts.md`). The single finding was gated by the evidence-mutation token (`scope=index_findings`) before persistence and sealed only after the (demo-automated) approval transition.

Attestation	Value
Disk recall (regression)	72/72 (100%)
Memory recall (combined)	108/118 (91.5%)
Test suite	4464
Evidence gate	PASS — finding persisted under single-scope mutation token (<code>scope=index_findings</code> , 30-min TTL); approval HMAC-sealed (demo automation, not human attested)

Every datum in this report traces to a real MCP capture or the sealed `report_generate(full)` snapshot for case `AMF-WIN-SAMPLE001` . No values were simulated. The lone approval was Playwright demo automation, explicitly flagged wherever it appears; a production close-out requires a human examiner HMAC sign-off.

13. Recommendations

1. [P1] Dump and disassemble the 10 RWX VADs in `winlogon.exe` (PID 628) — `vaddump` / `dumpfiles` followed by static triage — to confirm whether the injected regions are malicious shellcode or benign JIT/unpacked code, and to extract any embedded IOCs.
2. [P2] Investigate `wc.exe -e -o h.out` (PID 364, child of `svchost -k netsvcs`) — a non-standard binary with cryptic arguments spawned ~3.5 hrs after boot; recover the on-disk image and `h.out` output file, and validate against the `svchost` service host.
3. [P3] Replace the demo Playwright approval with a genuine human examiner HMAC sign-off before this case is treated as evidentiary, and capture a paired disk image if available to recover the initial-access vector that a RAM-only snapshot cannot show.

► Appendix — methodology, tool versions, chain of custody